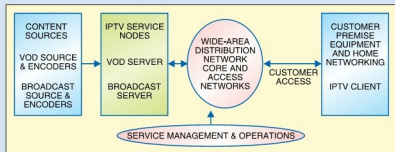


Building Blocks of IPTV

The blocks that constitute IPTV are:

1. Content sources
2. Service nodes
3. Wide area distribution networks
4. Customer access links
5. Customer premises equipment
6. IPTV client



Each block behaves like an individual functionality. The 'content source' receives video content from the producers and other sources. This content is then encoded and reserved in a database for VoD. Similarly, the service node receives video streams in different formats. The data received is reformatted and encapsulated for transmission with appropriate quality of service (QoS). This makes it ready for delivery to the customers.

The next block, 'wide area distribution network,' is composed of distribution capability, capacity and QoS. It also consists of capabilities such as multicast, which are reliable. Also, timely distribution of IPTV data streams from the service nodes to customers' premises is possible.

The next block is 'customer access links,' in which we require high-speed DSL technologies such as ADSL2+ and VDSL. Using these, services can be provided over the existing loop plant and through telephone lines to homes. Customers can also receive data through the fiber-to-the-curb (FTTC) and DSL technologies. FTTH can also be used for delivery to the customer. Customer premises equipment is a device that is located at the customer's premises. The functionality is broadband network termination.

Other functionalities that it provides are routing gateway, STB or home networking capabilities. The 'IPTV client' is a functionality, which stops the IPTV traffic at the customers' premises. IPTV is just a client, such as an STB, which conducts functional processing. Functional processing includes setting up the connection and QoS with the service node and decoding video streams, such as SDTV or a HDTV.

Difference between IPTV & Internet TV

There is very little difference between IPTV and Internet TV. IPTV refers to a closed and proprietary TV system. It is somewhat akin to today's cable TV. The content of IPTV is sent over secure IP channels, which can be controlled.

Internet TV, on the other hand, is an open and evolving framework, to which small and medium-sized video producers contribute. One can select from a list of content that is available. It is an open setup and anyone having an Internet connection can access the data.

—By Abhijit Paul Choudhury, technical correspondent, EY

video and other content-rich interactive services play a vital role. Service providers and leading global players are aggressively vying to capture the booming Indian market to deliver the triple play of voice, video and data

services in an integrated wireless context. Telcos in India are constantly upgrading their networks to offer IPTV in order to retain customers and develop new revenue streams.

IPTV had several limping starts

in the past decade. The video server technology was immature, the bandwidth available on access network was inadequate and the cost of customer premises equipment (CPE) was too high. "Those barriers are now only hurdles that can be avoided strategically. Building an IPTV network is within the range of the telcos' technical ability but it requires a dramatic change of culture," says Anil Prakash, secretary general, IPTV India Forum.

What is IPTV

IPTV delivers television programming to households via a broadband connection, using Internet protocols. Since it is in digital format, all analogue TVs require an IPTV set-top box (STB). IPTV is clubbed with other services like video-on-demand (VoD), voice-over IP (VOIP) or digital phone and Web access, collectively referred to as triple play. Triple play implies high-speed Internet and television/telephone service over a single broadband connection. With wireless, it is called quadruple play and grouped services (triple and quadruple) are called multi-play. Accordingly, IPTV comprises TV services provided over network, controlled by service operators such as telecom, cable or Internet service providers.

How it works

"Although IPTV is on Internet protocol, it is a service on TV and hence, does not require any computer or Internet bandwidth. It is different from the conventional analogue TV. IPTV runs through a telephone line and an STB. It is a real-time interactive medium by which a user request can be processed in real time. This feature allows users to watch different programmes depending on their choices and time availability," expatiates K.S. Choudhary, managing director, Aksh Optifibre, the first company to have launched IPTV services in India. The company is already providing IPTV and VOIP services in Delhi and Mumbai, under an agreement with MTNL.